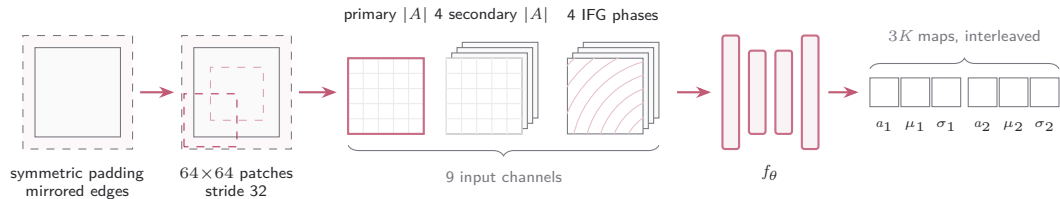


DLR-TomoSAR — Input & output representation

nine channels · two regimes · per-channel normalization

July 2026

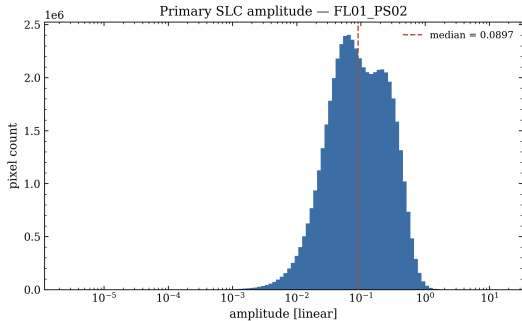
Input and output representation



- Scenes are symmetrically padded (edge-mirrored) so the 64×64 , stride-32 patch grid tiles them exactly; sliding-window stitching undoes the overlap at inference.
- The $3K$ output channels are interleaved per scatterer — $a_1, \mu_1, \sigma_1, a_2, \mu_2, \sigma_2, \dots$ — not grouped by parameter type.
- Outputs live in normalized space; denormalised and clamped to physical bounds before any curve is reconstructed.
- Current configuration $K=2$ (6 maps); the $K=5$ variant (15 maps) in development.

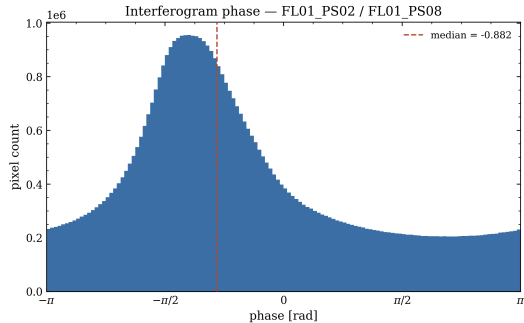
Two input channels, two regimes

primary amplitude



Log-scaled axis: the bulk sits near 10^{-1} but the tail runs past 10^0 — orders of magnitude in one channel. **Heavy-tailed** → robust log1p scaling.

interferogram phase



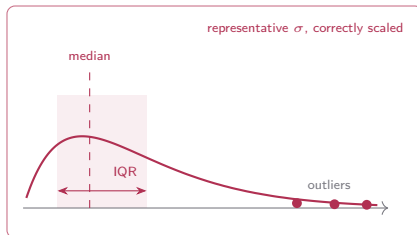
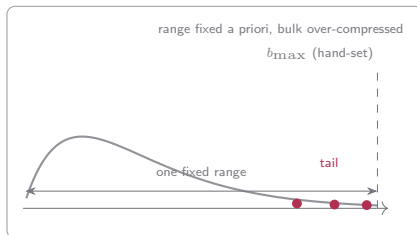
Confined to $[-\pi, \pi]$ by construction, a single broad mode — no tail to tame. **Bounded by physics** → fixed $\div \pi$.

Real F-SAR channels (primary PS02, IFG PS02/PS08). The nine inputs split cleanly into these two shapes — and the normalization strategy (next) follows the split.

Per-channel normalization — old → new, channel by channel

z-score — standard score · IQR — interquartile range

channel	old		new	why
pass / mag	log1p, unscaled	→	robust IQR (log1p)	heavy-tailed input
ifg / phase	fixed $[-\pi, \pi] \rightarrow [0, 1]$	→	fixed $\div \pi$	bounded by physics
out / a	hand-set min-max	→	robust IQR (log1p)	heavy-tailed target
out / σ	hand-set min-max	→	robust IQR (log1p)	heavy-tailed target
out / μ	hand-set min-max	→	z-score	bounded to sampling range



$$\tilde{x} = \frac{x - b_{\min}}{b_{\max} - b_{\min}}$$

old — fixed maps, nothing fitted. One hand-set range serves every scene; the tail owns the range and the bulk compresses toward zero.

$$\tilde{x} = \frac{\log(1+x) - \text{med}}{Q_3 - Q_1}$$

new — **quantiles fit the bulk**. The median and IQR ignore the tail; the bulk keeps its spread.

Normalization ablation — the inputs are flat, the output channels carry the gain

metric	base legacy fixed	+pass mag IQR log1p	+ifg φ $\div \pi$	+out amp IQR log1p	+out μ z-score	+out σ IQR log1p
curve R^2 $\uparrow$	0.58 $\pm$.01	0.55 $\pm$.02	0.56 $\pm$.01	0.59 $\pm$.02	0.59 $\pm$.01	0.57 $\pm$.02
curve MAE $\downarrow$	0.0537 $\pm$.002	0.0556 $\pm$.001	0.0539 $\pm$.0004	0.0504 $\pm$.001	0.0508 $\pm$.001	0.0501 $\pm$.001
PSNR [dB] $\uparrow$	50.4 $\pm$.1	50.1 $\pm$.2	50.3 $\pm$.1	50.5 $\pm$.2	50.6 $\pm$.1	50.4 $\pm$.2
per-pixel R^2 $\uparrow$	0.057 $\pm$.015	0.045 $\pm$.014	0.052 $\pm$.011	0.426 $\pm$.026	0.367 $\pm$.020	0.359 $\pm$.050
profile cos med $\uparrow$	0.000 $\pm$.000	0.000 $\pm$.000	0.000 $\pm$.000	0.890 $\pm$.012	0.872 $\pm$.021	0.851 $\pm$.020
SSIM elevation $\uparrow$	0.954 $\pm$.001	0.953 $\pm$.001	0.954 $\pm$.000	0.958 $\pm$.001	0.957 $\pm$.001	0.958 $\pm$.001
SSIM range $\uparrow$	0.907 $\pm$.002	0.906 $\pm$.001	0.907 $\pm$.001	0.927 $\pm$.002	0.924 $\pm$.001	0.926 $\pm$.003
SSIM azimuth $\uparrow$	0.935 $\pm$.002	0.934 $\pm$.001	0.935 $\pm$.001	0.945 $\pm$.001	0.944 $\pm$.001	0.946 $\pm$.001
matched F1 $\uparrow$	0.455 $\pm$.005	0.458 $\pm$.002	0.458 $\pm$.008	0.746 $\pm$.019	0.721 $\pm$.017	0.728 $\pm$.040
matched recall $\uparrow$	0.334 $\pm$.004	0.340 $\pm$.002	0.338 $\pm$.008	0.690 $\pm$.025	0.651 $\pm$.024	0.657 $\pm$.056
count exact $\uparrow$	0.228 $\pm$.007	0.231 $\pm$.005	0.232 $\pm$.008	0.655 $\pm$.033	0.599 $\pm$.030	0.613 $\pm$.066
peak err mean $\downarrow$	11.2 $\pm$.1	11.0 $\pm$.1	11.1 $\pm$.1	4.0 $\pm$.2	4.6 $\pm$.3	4.5 $\pm$.9
median $\downarrow$	14.1 $\pm$.0	14.1 $\pm$.0	14.1 $\pm$.0	0.7 $\pm$.0	0.7 $\pm$.0	0.9 $\pm$.4
p95 $\downarrow$	16.8 $\pm$.0	16.8 $\pm$.0	16.8 $\pm$.0	14.9 $\pm$.3	15.6 $\pm$.3	15.4 $\pm$.7
matched μ MAE $\downarrow$	3.33 $\pm$.05	3.39 $\pm$.04	3.36 $\pm$.04	2.24 $\pm$.08	2.29 $\pm$.07	2.30 $\pm$.12
matched σ MAE $\downarrow$	1.35 $\pm$.02	1.36 $\pm$.01	1.35 $\pm$.02	0.87 $\pm$.04	0.92 $\pm$.04	0.86 $\pm$.05
matched a MAE $\downarrow$	0.547 $\pm$.015	0.542 $\pm$.023	0.534 $\pm$.015	0.321 $\pm$.011	0.364 $\pm$.024	0.363 $\pm$.034

UNet · sorted-GT · $K=2$ · param-MSE · no aug · per-slot · patch 64×32 ; held-out validation band. Cells: mean $\pm$ seed std over five seeded replicas per rung. **bold** = best in row, marked only where best and worst differ by more than 5%; **red heads** = the output rungs; **green** = the metrics the out-amp rung unlocks. Base = the legacy fixed maps: unscaled log1p pass mag, angle $\rightarrow [0, 1]$ ifg φ , hand-set min-max outputs; each rung switches one channel group to its final strategy and keeps it. The three output arms differ within seed noise (paired $p \geq 0.31$). μ , σ , peak in elevation units; a in linear amplitude. Peak median/p95: per-pixel peak-error distribution; profile cos med: median per-pixel cosine to the GT profile.

Normalization ablation — the gains concentrate on single-scatterer pixels

metric	base legacy fixed	+pass mag IQR log1p	+ifg φ $\div \pi$	+out amp IQR log1p	+out μ z-score	+out σ IQR log1p
matched precision $\uparrow$	0.711 $\pm$.013	0.704 $\pm$.003	0.710 $\pm$.006	0.814 $\pm$.011	0.808 $\pm$.010	0.816 $\pm$.015
$k=1$	0.588 $\pm$.023	0.585 $\pm$.013	0.592 $\pm$.018	0.853 $\pm$.016	0.835 $\pm$.013	0.848 $\pm$.019
$k=2$	0.772 $\pm$.009	0.765 $\pm$.005	0.769 $\pm$.003	0.761 $\pm$.009	0.774 $\pm$.007	0.776 $\pm$.013
matched recall $\uparrow$	0.334 $\pm$.004	0.340 $\pm$.002	0.338 $\pm$.008	0.690 $\pm$.025	0.651 $\pm$.024	0.657 $\pm$.056
$k=1$	0.156 $\pm$.005	0.163 $\pm$.006	0.161 $\pm$.007	0.707 $\pm$.041	0.641 $\pm$.042	0.650 $\pm$.088
$k=2$	0.586 $\pm$.007	0.590 $\pm$.004	0.590 $\pm$.009	0.664 $\pm$.009	0.666 $\pm$.006	0.668 $\pm$.016
count acc pred $k=1$ $\uparrow$	0.752 $\pm$.015	0.716 $\pm$.018	0.730 $\pm$.024	0.961 $\pm$.003	0.955 $\pm$.003	0.952 $\pm$.009
count acc pred $k=2$ $\uparrow$	0.705 $\pm$.017	0.697 $\pm$.008	0.704 $\pm$.012	0.712 $\pm$.017	0.702 $\pm$.014	0.720 $\pm$.013
matched μ MAE $\downarrow$	3.33 $\pm$.05	3.39 $\pm$.04	3.36 $\pm$.04	2.24 $\pm$.08	2.29 $\pm$.07	2.30 $\pm$.12
$k=1$	1.03 $\pm$.08	1.11 $\pm$.04	1.05 $\pm$.02	0.55 $\pm$.03	0.59 $\pm$.07	0.63 $\pm$.06
$k=2$	4.01 $\pm$.06	4.07 $\pm$.04	4.05 $\pm$.04	4.18 $\pm$.07	4.09 $\pm$.06	4.07 $\pm$.05
matched σ MAE $\downarrow$	1.35 $\pm$.02	1.36 $\pm$.01	1.35 $\pm$.02	0.87 $\pm$.04	0.92 $\pm$.04	0.86 $\pm$.05
$k=1$	0.60 $\pm$.01	0.60 $\pm$.01	0.60 $\pm$.01	0.24 $\pm$.02	0.27 $\pm$.02	0.25 $\pm$.02
$k=2$	1.57 $\pm$.02	1.59 $\pm$.01	1.58 $\pm$.02	1.59 $\pm$.01	1.62 $\pm$.03	1.51 $\pm$.01
matched a MAE $\downarrow$	0.547 $\pm$.015	0.542 $\pm$.023	0.534 $\pm$.015	0.321 $\pm$.011	0.364 $\pm$.024	0.363 $\pm$.034
$k=1$	0.433 $\pm$.029	0.413 $\pm$.035	0.404 $\pm$.032	0.126 $\pm$.009	0.153 $\pm$.023	0.155 $\pm$.029
$k=2$	0.580 $\pm$.011	0.581 $\pm$.019	0.574 $\pm$.011	0.545 $\pm$.016	0.588 $\pm$.016	0.586 $\pm$.035

UNet · sorted-GT · $K=2$ · param-MSE · no aug · per-slot · patch 64×32 ; held-out validation band. Cells: mean $\pm$ seed std over five seeded replicas; k = true scatterer count of the pixel. **bold** = best in row, marked only where best and worst differ by more than 5%; **red heads** = the output rungs; **green** = the $k=1$ collapse they buy, **red** = the one cost, second-scatterer μ . Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k . μ , σ in elevation units; a in linear amplitude.

Normalization ablation — slot 0 calibrates, slot 1 trades leakage against amplitude

prediction mean	GT	base legacy fixed	+pass mag IQR log1p	+ifg φ $\div \pi$	+out amp IQR log1p	+out μ z-score	+out σ IQR log1p
<i>slot 0 — strongest scatterer; GT-active on every pixel</i>							
active frac	1.00	0.317 $\pm$.006	0.326 $\pm$.004	0.323 $\pm$.009	0.754 $\pm$.030	0.702 $\pm$.030	0.708 $\pm$.066
a — active	0.77	2.01 $\pm$.06	2.01 $\pm$.06	2.03 $\pm$.03	0.86 $\pm$.04	0.89 $\pm$.04	0.89 $\pm$.07
μ — active	-4.89	-3.47 $\pm$.12	-3.60 $\pm$.12	-3.46 $\pm$.10	-4.28 $\pm$.09	-4.16 $\pm$.11	-4.03 $\pm$.15
σ — active	1.76	2.46 $\pm$.04	2.41 $\pm$.07	2.42 $\pm$.06	1.61 $\pm$.06	1.69 $\pm$.09	1.52 $\pm$.07
<i>slot 1 — second scatterer; GT-active on 26% of pixels</i>							
active frac	0.261	0.276 $\pm$.009	0.282 $\pm$.003	0.279 $\pm$.008	0.315 $\pm$.009	0.314 $\pm$.007	0.307 $\pm$.008
a — active	1.42	1.43 $\pm$.03	1.40 $\pm$.04	1.42 $\pm$.02	1.07 $\pm$.03	1.04 $\pm$.01	1.06 $\pm$.05
a — inact.	0.00	0.74 $\pm$.03	0.80 $\pm$.02	0.76 $\pm$.05	0.44 $\pm$.02	0.45 $\pm$.03	0.45 $\pm$.04
μ — active	11.39	10.95 $\pm$.26	10.39 $\pm$.42	10.79 $\pm$.29	9.95 $\pm$.33	10.18 $\pm$.24	9.97 $\pm$.40
μ — inact.	—	4.29 $\pm$.17	3.76 $\pm$.37	4.26 $\pm$.22	3.46 $\pm$.69	3.17 $\pm$.21	3.39 $\pm$.57
σ — active	3.55	3.70 $\pm$.09	3.65 $\pm$.08	3.66 $\pm$.06	3.38 $\pm$.11	3.55 $\pm$.08	3.02 $\pm$.04
σ — inact.	—	2.30 $\pm$.09	2.36 $\pm$.07	2.30 $\pm$.04	1.99 $\pm$.09	2.19 $\pm$.09	1.90 $\pm$.04
<i>distribution width — std over GT-active pixels</i>							
a — slot 0	1.48	1.43 $\pm$.07	1.45 $\pm$.07	1.48 $\pm$.04	1.23 $\pm$.04	1.18 $\pm$.05	1.17 $\pm$.06
μ — slot 0	3.30	2.58 $\pm$.11	2.46 $\pm$.08	2.54 $\pm$.10	1.99 $\pm$.08	2.08 $\pm$.14	2.03 $\pm$.12
σ — slot 0	2.84	1.22 $\pm$.05	1.08 $\pm$.07	1.17 $\pm$.06	1.26 $\pm$.06	1.29 $\pm$.06	1.01 $\pm$.03
a — slot 1	1.45	0.97 $\pm$.05	0.94 $\pm$.04	0.98 $\pm$.01	0.87 $\pm$.05	0.80 $\pm$.03	0.84 $\pm$.05
μ — slot 1	16.30	7.57 $\pm$.22	7.29 $\pm$.25	7.45 $\pm$.09	8.09 $\pm$.20	8.87 $\pm$.27	8.76 $\pm$.27
σ — slot 1	4.11	1.17 $\pm$.07	1.12 $\pm$.05	1.17 $\pm$.03	1.28 $\pm$.03	1.28 $\pm$.04	1.14 $\pm$.04

UNet · sorted-GT · $K=2$ · param-MSE · no aug · per-slot · patch 64×32 ; held-out validation band. Cells: mean $\pm$ seed std over five seeded replicas; means conditioned on the slot's GT activity mask. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%; **green** = the calibration shifts the out-amp rung buys, **red** = the one overshoot past the GT. Inactive GT slots define no μ/σ reference (—). Width rows: std of the prediction over GT-active pixels vs the GT's own std. μ, σ in elevation units; a in linear amplitude.